

Olist E-Commerce Analysis

Customer Satisfaction & Operational Risk

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Business Analyst | Entry Level Project

8.1%

Late Delivery Rate

11.9%

1-Star Review Rate

0.87

Correlation: Late → 1★

Python

Excel

PowerPoint

Word

Root Cause Analysis

Risk Register

Process Mapping

PROJECT OVERVIEW

This case study presents a complete end-to-end Business Analysis of the **Olist Brazilian E-Commerce dataset (2016–2018)**, covering 112,650 customer orders. The objective was to identify the root causes of customer dissatisfaction, quantify business risk, and deliver strategic recommendations, demonstrating core BA skills including data analysis, process modelling, risk assessment, and stakeholder communication.

THE PROBLEM

Olist is a third-party marketplace connecting Brazilian sellers to customers nationwide. Initial exploration revealed a significant polarisation in review scores. The analytical challenge was to determine whether late deliveries were the *only* driver of dissatisfaction, or whether other independent variables were at play.

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Late Delivery Rate

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KEY FINDINGS

01 Late Deliveries → 1-Star Reviews

Pearson correlation 0.87 statistically strong. 8.1% of all orders arrived late, directly driving the 11.9% 1-star rate.

02 Regional Logistics Gap

States AL, MA, PI significantly underperform nationally. A structural carrier network gap requiring a regional strategy.

03 Product Quality: Independent Driver

Office Furniture: 20.5% 1-star rate on ON-TIME deliveries. Seller-side quality is a separate failure point from logistics.

ROOT CAUSE ANALYSIS

The analysis identified **two independent failure points**, a distinction that is critical to the strategic response:

1. Logistics failure, a 0.87 correlation between late delivery and 1-star reviews, with regional carrier gaps in AL, MA, and PI. Requires carrier governance and SLA enforcement.

2. Seller-side product quality, Office Furniture's 20.5% 1-star rate on on-time deliveries confirms that even when logistics perform correctly, product issues cause dissatisfaction. Requires seller accountability frameworks.

A single-variable solution, fixing deliveries only, would not resolve the full 11.9% 1-star review rate. A dual-track approach is required.

RISK ASSESSMENT

Four risks were identified and rated. The most significant is a **Critical-rated Marketplace Supply Risk**, the tension between tightening quality controls and maintaining catalogue breadth. This highlights the strategic trade-off inherent in scaling a third-party marketplace.

Risk	Likelihood	Impact	Rating
Delivery & Logistics Risk	Medium	High	High
Product Quality Risk	Medium	High	High
Marketplace Supply Risk	High	High	Critical
Change Implementation Risk	Low	High	Medium

STRATEGIC RECOMMENDATIONS

Implement Regional Carrier SLAs. Prioritise AL, MA, and PI for carrier performance review. Set delivery accuracy KPIs with an escalation trigger at 7% late rate threshold, directly targeting the 0.87 correlation.

Introduce Seller Quality Scorecards. Develop a seller-facing quality framework flagging high-risk categories. Require photo verification for any category exceeding 15% 1-star rate on on-time orders.

Phase Quality Controls to Protect Supply. Phase controls in gradually with seller support programmes, monitoring listing volume monthly before tightening compliance.

Pilot Changes Before National Rollout. Test carrier routing changes in a low-risk region first. Define rollback criteria and monitor review scores weekly during transition.

PROJECT DELIVERABLES

Deliverable		Format	Description
Business Doc	Requirements	Word (.docx)	Problem scope, objectives, stakeholder needs, success criteria
Excel Dashboard		Excel (.xlsx)	3 charts: State analysis, product category, late vs on-time reviews

Process Flow Diagram	PowerPoint (.pptx)	End-to-end order fulfillment with decision points & failure markers
Risk Register	Excel (.xlsx)	4 risks with likelihood, impact, rating, mitigation & ownership
Executive Summary	Word (.docx)	Data-driven strategic summary for stakeholder presentation
Portfolio Case Study	PDF	This document, full project narrative for hiring managers

SKILLS DEMONSTRATED

Data Analysis	Correlation modelling, segmentation, root cause analysis using Python & Excel
Requirements Doc	BRD covering problem scope, stakeholder needs, and success criteria
Process Modelling	End-to-end process flow with decision points and failure mapping
Risk Management	Structured risk register with likelihood/impact scoring & mitigations
Stakeholder Comms	Executive Summary written for senior stakeholders in strategic language
Strategic Thinking	Identified the tension between quality control and marketplace supply

Data source: Olist Brazilian E-Commerce public dataset (Kaggle, 2018). All findings are data-driven and reproducible.